

JEE MAIN REVISION MASTER: D AND F BLOCK ELEMENTS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 28 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Transition Elements: Properties, Oxidation States & Color	22 PYQs	d-Block Chemistry
Moderate	Potassium Dichromate, Permanganate & Lanthanoid Contraction	6 PYQs	f-Block & Compounds

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Transition Elements: Properties, Oxidation States & Color	$\mu_{\text{spin}} = \sqrt{n(n+2)} \text{ BM}$, $\text{Colorless: } d^0 \text{ and } d^{10} \text{ ions}$
Potassium Dichromate, Permanganate & Lanthanoid Contraction	$\text{Dichromate-Chromate Equilibrium: } \text{Cr}_2\text{O}_7^{2-} \rightleftharpoons 2\text{CrO}_4^{2-} + 2\text{H}^+ + 2\text{OH}^-$ (Orange) (Yellow)

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Transition Elements: Properties, Oxidation States & Color (Very High Yield — 22 PYQs)

Core Principle: Transition metals feature incompletely filled d-orbitals. Magnetic moment $\mu = \sqrt{n(n+2)} \text{ BM}$ (n is number of unpaired electrons; Mn^{2+} with $3d^5$ has maximum $\mu = \sqrt{35} \approx 5.92 \text{ BM}$). Color arises from d-d transitions (ions with d^0 like Sc^{3+} , Ti^{4+} or d^{10} like Zn^{2+} are completely colorless). Highest oxidation state +7 is exhibited by Manganese (KMnO_4).

■ **Caution / JEE Pitfall:** Zinc, Cadmium, and Mercury are NOT considered true transition elements because they possess completely filled d^{10} configurations both in elemental and ionic states.

■ **Speed Trick:** Interstitial compounds (e.g. TiC , Fe_3H) retain metallic conductivity, have very high melting points, and are chemically inert and extremely hard.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The calculated spin only magnetic moment of Cr^{2+} ion is..." [Ans: B]
- [2026 Shift] Q#2 (MCQ): "The highest oxidation state of manganese in fluoride is +4 (MnF_4), but the highest oxidation state in oxides is +7 (Mn_2O_7)" [Ans: D]
- [2025 Shift] Q#1 (MCQ): "A member of the Lanthanoid series which is well known to exhibit +4 oxidation state is..." [Ans: N/A]

3.2 Potassium Dichromate, Permanganate & Lanthanoid Contraction (Moderate Yield — 6 PYQs)

Core Principle: $\text{K}_2\text{Cr}_2\text{O}_7$: Orange dichromate $\text{Cr}_2\text{O}_7^{2-}$ interconverts to yellow chromate CrO_4^{2-} in alkaline medium: $\text{Cr}_2\text{O}_7^{2-} + 2\text{OH}^- \rightleftharpoons 2\text{CrO}_4^{2-} + \text{H}_2\text{O}$. Chromyl chloride test: CrO_2Cl_2 red vapors confirm ionic chloride. Lanthanoid contraction: Steady decrease in atomic/ionic radii due to poor shielding by 4f electrons, making pairs like Zr (4d) and Hf (5d) have almost identical atomic radii (160 pm vs 159 pm).

■■ **Caution / JEE Pitfall:** Dichromate ion $\text{Cr}_2\text{O}_7^{2-}$ and permanganate MnO_4^- are intensely colored despite having d^0 configurations; their color is due to CHARGE TRANSFER spectra, NOT d-d transitions!

■ **Speed Trick:** Lanthanoid contraction causes Zr and Hf to have virtually identical chemical properties, making their separation extremely difficult.

Representative Source PYQs:

- [2026 Shift] Q#3 (MCQ): "Which of the following will not act as an oxidising agent?..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "Which of the following statements related to lanthanoids is incorrect?..." [Ans: N/A]
- [2022 Shift] Q#2 (MCQ): "The only lanthanoid which is radioactive..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The calculated spin only magnetic moment of Cr^{2+} ion is

- (A) 3.87 BM
- (B) 4.90 BM [Correct]
- (C) 5.92 BM
- (D) 2.84 BM

Correct Answer: B

Solution Analysis: For spin only magnetic moment, we use the formula $\mu = \sqrt{n(n+2)}\text{BM}$ where n is the number of unpaired electrons. Chromium has atomic number 24. Its electronic configuration is $\text{Cr} = [\text{Ar}]3d^5 4s^1$. For Cr^{2+} , two electrons are removed. First the 4s electron is removed, then one electron from 3d. So, $\text{Cr}^{2+} = [\text{Ar}]3d^4$. In $3d^4$, according to Hund's rule, the four electrons occupy four different d orbitals singly. So the number of unpaired electrons is $n=4$. Now, $\mu = \sqrt{4(4+2)} = \sqrt{24}$ $\mu = 4.90\text{BM}$. So the correct answer is B.

Benchmark Problem 2: KCET 2026 (Q#3)

Which of the following will not act as an oxidising agent?

- (A) CrO_3
- (B) MoO_3
- (C) CrO_4^{2-}
- (D) $\text{Cr}_2\text{O}_7^{2-}$

Correct Answer: None

Solution Analysis: MoO_3 will not act as an oxidising agent. So, the correct option is: Option B MoO_3 . In all the given species, the metal is in the +6 oxidation state. For chromium: CrO_3 , CrO_4^{2-} , $\text{Cr}_2\text{O}_7^{2-}$ chromium is in the +6 oxidation state. Chromium in +6 state is easily reduced to Cr^{3+} , especially in acidic medium. So these compounds behave as strong oxidising agents. For example, dichromate ion gets reduced as: $\text{Cr}_2\text{O}_7^{2-} \rightarrow \text{Cr}^{3+}$. Hence, CrO_3 , CrO_4^{2-} , $\text{Cr}_2\text{O}_7^{2-}$ all act as oxidising agents.